

1. Simple Interest and Compound Interest:

(a) *Simple Interest* = $SI = \frac{prt}{100}$, where p = Principal, r = Rate of Interest, t = Time .

(b) *Rate* = $r = \frac{100 \times SI}{p \times t}$.

(c) *Principal* = $p = \frac{100 \times SI}{r \times t}$.

(d) *Time* = $t = \frac{100 \times SI}{p \times r}$.

(e) *Amount* = Principal + Simple Interest .

(f) *Compound Interest* = $p \left\{ \left(1 + \frac{r}{100} \right)^t - 1 \right\}$. Here interest compounded annually.

(g) *Amount* = $p \left(1 + \frac{r}{100} \right)^t$. Here interest compounded annually.

(h) *Compound Interest* = $p \left\{ \left(1 + \frac{r}{200} \right)^{2t} - 1 \right\}$. Here interest compounded half yearly.

(i) *Amount* = $p \left(1 + \frac{r}{200} \right)^{2t}$. Here interest compounded half yearly.

(j) *Compound Interest* = $p \left\{ \left(1 + \frac{r}{400} \right)^{4t} - 1 \right\}$. Here interest compounded Quarterly.

(k) *Amount* = $p \left(1 + \frac{r}{400} \right)^{4t}$. Here interest compounded Quarterly.

(l) If rate of Interest for three consecutive years are r_1, r_2, r_3 , then

$$\text{Amount} = p \left(1 + \frac{r_1}{100} \right) \left(1 + \frac{r_2}{100} \right) \left(1 + \frac{r_3}{100} \right).$$

2. Profit & Loss:

(a) **Cost Price:** The price, at which an article is purchased, is called its cost price, abbreviated as C.P.

(b) **Selling Price:** The price, at which an article is sold, is called its selling prices, abbreviated as S.P.

(c) **Profit or Gain:** If S.P. is greater than C.P., the seller is said to have a profit or gain.

(d) **Loss:** If S.P. is less than C.P., the seller is said to have incurred a loss.

(e) *Gain* = Selling Price + Cost Price .

(f) *Loss* = Selling Price – Cost Price .

(g) *Cost Price* = Selling Price – Gain = Selling Price + Loss .

(h) *Selling Price* = Cost Price + Gain = Cost Price – Loss .

$$(i) \text{ Selling Price} = \frac{100 + \text{Gain}\%}{100} \times \text{Cost Price} = \frac{100 - \text{Loss}\%}{100} \times \text{Cost Price}.$$

$$(j) \text{ Cost Price} = \frac{100}{100 + \text{Gain}\%} \times \text{Selling Price} = \frac{100}{100 - \text{Loss}\%} \times \text{Selling Price}.$$

$$(k) \text{ Gain}\% = \frac{\text{Gain}}{\text{Cost Price}} \times 100\%.$$

$$(l) \text{ Loss}\% = \frac{\text{Loss}}{\text{Cost Price}} \times 100\%$$

(m) If an article is sold at a gain of say 35%, then S.P. = 135% of C.P.

(n) If an article is sold at a loss of say, 35% then S.P. = 65% of C.P.

(o) When a person sells two similar items, one at a gain of say $x\%$, and the other at a loss of $x\%$, then the seller always incurs a loss given by: $\text{Loss}\% = \frac{\text{Common Loss and Gain}\%}{10} \times 2.$

(p) If a trader professes to sell his goods at cost price, but uses false weights, then

$$\text{Gain}\% = \frac{\text{Error}}{\text{True Value} - \text{Error}} \times 100\%$$

3. **Percentage:**

(a) By a certain percent, we mean that many hundredths.

(b) Thus, x percent means x hundredths, written as $x\%$.

(c) To express $x\%$ as a fraction: We have, $x\% = \frac{x}{100}.$

(d) To Express $\frac{a}{b}$ as a percent, we have $\frac{a}{b} = \left(\frac{a}{b} \times 100 \right) \%$.

(e) If the price of a commodity increases by $R\%$, then the reduction in consumption so as not to increase the expenditure is: $\left(\frac{R}{100 + R} \times 100 \right) \%$.

(f) If the price of a commodity decreases by $R\%$, then the increase in consumption so as not to decrease the expenditure is: $\left(\frac{R}{100 - R} \times 100 \right) \%$.

(g) **Population:** Let the population of a town be P now and suppose it increases at the rate of $R\%$ per annum, then:

$$\text{Population after } n \text{ years} = P \left(1 + \frac{R}{100} \right)^n.$$

$$\text{Population } n \text{ years ago} = \frac{P}{\left(1 + \frac{R}{100} \right)^n}.$$

(h) **Depreciation:** Let the present value of a machine be P . Suppose it depreciates at the rate of $R\%$ per annum. Then:

$$\text{Value of the machine after } n \text{ years} = P \left(1 - \frac{R}{100} \right)^n$$

$$\text{Value of the machine } n \text{ years ago} = \frac{P}{\left(1 - \frac{R}{100}\right)^n}.$$

- (i) If A is R% more than B, then B is less than A by $\left(\frac{R}{100+R} \times 100\right)\%$.
- (j) If A is R% less than B, then B is more than A by $\left(\frac{R}{100-R} \times 100\right)\%$.

4. Work & Time:

- (a) If A finish a piece of work in n days, the work done by A in one day = $\frac{1}{n}$ work.
- (b) If A's one day work = $\frac{1}{n}$, then the time taken to finish the work by A = n days.
- (c) **Ratio:**
 If A is thrice as good a workman as B, then:
 Ratio of work done by A and B = 3 : 1.
 Ratio of times taken by A and B to finish a work = 1 : 3.
- (d) If A can do a piece of work in p days, B can do the same work in q days, then together A and B can do the work in $\frac{pq}{p+q}$ days.
- (e) If A can do a piece of work in p days, B can do the same work in q days, and C can do the work in r days, then together A, B and C can do the work in $\frac{pqr}{pq+qr+rp}$ days.
- (f) If A & B can do a piece of work in p days, B & C can do the same work in q days, and C & A can do the work in r days, then together A, B and C can do the work in $\frac{2pqr}{pq+qr+rp}$ days.
- (g) If M_1 men can do W_1 work in D_1 days working H_1 hours per day and M_2 men can do W_2 work in D_2 days working H_2 hours per day, then $\frac{M_1 D_1 H_1}{W_1} = \frac{M_2 D_2 H_2}{W_2}$.

5. Distance, Speed & Time (Train):

- (a) $\text{Speed} = \frac{\text{Distance}}{\text{Time}}$.
- (b) $\text{Time} = \frac{\text{Distance}}{\text{Speed}}$.
- (c) $\text{Distance} = \text{Speed} \times \text{Time}$.
- (d) **km/hr to m/sec conversion:** $x \text{ km/hr} = x \times \frac{5}{18} \text{ m/sec}$.
- (e) **m/sec to km/hr conversion:** $x \text{ m/sec} = x \times \frac{18}{5} \text{ km/hr}$.
- (f) Time taken by a train of length l metres to pass a pole or standing man or a signal post is equal to the time taken by the train to cover l metres.

- (g) Time taken by a train of length l metres to pass a stationary object of length b metres is the time taken by the train to cover $l + b$ metres.
- (h) Suppose two trains or two objects bodies are moving in the same direction at u m/s and v m/s, where $u > v$, then their relative speed is $= (u - v)$ m/s.
- (i) Suppose two trains or two objects bodies are moving in opposite directions at u m/s and v m/s, then their relative speed is $= (u + v)$ m/s.
- (j) If two trains of length a metres and b metres are moving in opposite directions at u m/s and v m/s, then: The time taken by the trains to cross each other $= \frac{a+b}{u+v}$ sec
- (k) If two trains of length a metres and b metres are moving in the same direction at u m/s and v m/s, then: The time taken by the faster train to cross the slower train $= \frac{a+b}{u-v}$ sec.
- (l) If two trains (or bodies) start at the same time from points A and B towards each other and after crossing they take a and b sec in reaching B and A respectively, then:

$$\frac{A's \text{ Speed}}{B's \text{ Speed}} = \frac{\sqrt{b}}{\sqrt{a}}.$$
- (m) Suppose a man covers a certain distance at x km/hr and an equal distance at y km/hr. Then, the average speed during the whole journey is $\frac{2xy}{x+y}$ km/hr.
- (n) If the ratio of the speeds of A and B is $a : b$, then the ratio of the times taken by them to cover the same distance is $\frac{1}{a} : \frac{1}{b} = b : a$.

6. Pipes and Cistern:

- (a) **Inlet:** A pipe connected with a tank or a cistern or a reservoir, that fills it, is known as an inlet.
- (b) **Outlet:** A pipe connected with a tank or cistern or reservoir, emptying it, is known as an outlet.
- (c) If a pipe can fill a tank in x hours, then: part can be filled in 1 hr $= \frac{1}{x}$.
- (d) If a pipe can empty a tank in y hours, then: part emptied in 1 hr $= \frac{1}{y}$.
- (e) If a pipe can fill a tank in x hours and another pipe can empty the full tank in y hours (where $y > x$), then on opening both the pipes, then the net part filled in 1 hour $= \frac{1}{x} - \frac{1}{y}$.
- (f) If a pipe can fill a tank in x hours and another pipe can empty the full tank in y hours (where $x > y$), then on opening both the pipes, then the net part emptied in 1 hour $= \frac{1}{y} - \frac{1}{x}$.

7. Partnership:

- (a) **Partnership:** When two or more than two persons run a business jointly, they are called *partners* and the deal is known as *partnership*.
- (b) **Working and Sleeping Partners:** A partner who manages the business is known as a *working partner* and the one who simply invests the money is a *sleeping partner*.

(c) Ratio of Divisions of Gains:

- (i) When investments of all the partners are for the same time, the gain or loss is distributed among the partners in the ratio of their investments.

Suppose A and B invest Rs. x and Rs. y respectively for a year in a business, then at the end of the year:

(A's share of profit) : (B's share of profit) = $x : y$.

- (ii) When investments are for different time periods, then equivalent capitals are calculated for a unit of time by taking (capital $\times$ number of units of time). Now gain or loss is divided in the ratio of these capitals.

Suppose A invests Rs. x for p months and B invests Rs. y for q months then,

(A's share of profit) : (B's share of profit) = $xp : yq$.

8. Variation:

- (a) Direct Variation: Two quantities are said to be directly proportional, if on the increase (or decrease) of the one, the other increases (or decreases) to the same extent. Two quantities a and b

are said to vary directly if $\frac{a}{b} = \text{constant}$.

- (b) Indirect Variation: Two quantities are said to be indirectly proportional, if on the increase of the one, the other decreases to the same extent and vice-versa. Two quantities a and b said to vary indirectly if $ab = \text{constant}$.

- (c) Mixed Variation: If $x \propto y(z \text{ constant})$ and $x \propto z(y \text{ constant})$, mixed variation when y and z

vary then $\frac{x_1}{x_2} = \frac{y_1}{y_2} \times \frac{z_1}{z_2}$.

9. BLOOD RELATION

The questions which are asked in this section depend upon *Relation*. You should have a sound knowledge of the blood relation in order to solve the questions.

To remember easily the relations may be divided into two sides as given below:

1. Relations of Paternal side:

1. Father's father $\rightarrow$ Grandfather
2. Father's mother $\rightarrow$ Grandmother
3. Father's brother $\rightarrow$ Uncle
4. Father's sister $\rightarrow$ Aunt
5. Children of uncle $\rightarrow$ Cousin
6. Wife of uncle $\rightarrow$ Aunt
7. Children of aunt $\rightarrow$ Cousin
8. Husband of aunt $\rightarrow$ Uncle

2. Relations of Maternal side:

1. Mother's father → Maternal grandfather
2. Mother's mother → Maternal grandmother
3. Mother's brother Maternal uncle
4. Mother's sister → Aunt
5. Children of maternal uncle → Cousin
6. Wife of maternal uncle → Maternal aunt

10. CALENDER**1. Odd Days:**

We are supposed to find the day of the week on a given date.

For this, we use the concept of 'odd days'.

In a given period, the number of days more than the complete weeks are called odd days.

2. Leap Year: Note: A leap year has 366 days.

(i) Every year divisible by 4 is a leap year, if it is not a century.

(ii) Every 4th century is a leap year and no other century is a leap year.

Examples:

- i. Each of the years 1948, 2004, 1676 etc. is a leap year.
- ii. Each of the years 400, 800, 1200, 1600, 2000 etc. is a leap year.
- iii. None of the years 2001, 2002, 2003, 2005, 1800, 2100 is a leap year.

3. Ordinary Year:

The year which is not a leap year is called an ordinary years. An ordinary year has 365 days.

4. Counting of Odd Days:

1. 1 ordinary year = 365 days = (52 weeks + 1 day.)

∴ 1 ordinary year has 1 odd day.

2. 1 leap year = 366 days = (52 weeks + 2 days)

∴ 1 leap year has 2 odd days.

3. 100 years = 76 ordinary years + 24 leap years

= (76 x 1 + 24 x 2) odd days = 124 odd days.

= (17 weeks + days) = 5 odd days.

∴ Number of odd days in 100 years = 5.

Number of odd days in 200 years = (5 x 2) = 3 odd days.

Number of odd days in 300 years = (5 x 3) = 1 odd day.

Number of odd days in 400 years = (5 x 4 + 1) = 0 odd day.

Similarly, each one of 800 years, 1200 years, 1600 years, 2000 years etc. has 0 odd days.

5. Day of the Week Related to Odd Days:

No. of days	0	1	2	3	4	5	6
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Day	Sun	Mon	Tue	Wed	Thu	Fri	Sat
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11. Stocks & Shares

1. **Stock Capital:**
The total amount of money needed to run the company is called the stock capital.
2. **Shares or Stock:**
The whole capital is divided into small units, called shares or stock.
For each investment, the company issues a 'share-certificate', showing the value of each share and the number of shares held by a person.
The person who subscribes in shares or stock is called a share holder or stock holder.
3. **Dividend:**
The annual profit distributed among share holders is called dividend.
Dividend is paid annually as per share or as a percentage.
4. **Face Value:**
The value of a share or stock printed on the share-certificate is called its Face Value or Nominal Value or Par Value.
5. **Market Value:**
The stock of different companies are sold and bought in the open market through brokers at stock-exchanges. A share or stock is said to be:
 - i. At premium or Above par, if its market value is more than its face value.
 - ii. At par, if its market value is the same as its face value.
 - iii. At discount or Below par, if its market value is less than its face value.
 Thus, if a Rs. 100 stock is quoted at premium of 16,
 then market value of the stock = Rs. (100 + 16) = Rs. 116.
 Likewise, if a Rs. 100 stock is quoted at a discount of 7,
 then market value of the stock = Rs. (100 - 7) = 93.
6. **Brokerage:**
The broker's charge is called brokerage.
 - (i) When stock is purchased, brokerage is added to the cost price.
 - (ii) When stock is sold, brokerage is subtracted from the selling price.
 Remember:
 - i. The face value of a share always remains the same.
 - ii. The market value of a share changes from time to time.
 - iii. Dividend is always paid on the face value of a share.
 - iv. Number of shares held by a person

$$\frac{\text{Total Investment}}{\text{Investment in 1 share}} = \frac{\text{Total Income}}{\text{Income from 1 share}} = \frac{\text{Total Face Value}}{\text{Face Value of 1 share}}$$
7. Thus, by a Rs. 100, 9% stock at 120, we mean that:
 - i. Face Value of stock = Rs. 100.
 - ii. Market Value (M.V) of stock = Rs. 120.
 - iii. Annual dividend on 1 share = 9% of face value = 9% of Rs. 100 = Rs. 9.
 - iv. An investment of Rs. 120 gives an annual income of Rs. 9.
 - v. Rate of interest p.a = Annual income from an investment of Rs. 100

$$\frac{9}{120} \times 100\% = 7.5\%$$

12. Banker's Discount

Banker's Discount: Suppose a merchant A buys goods worth, say Rs. 10,000 from another merchant B at a credit of say 5 months. Then, B prepares a bill, called the bill of exchange. A signs this bill and allows B to withdraw the amount from his bank account after exactly 5 months.

The date exactly after 5 months is called **nominally due date**. Three days (known as grace days) are added to it get a date, known as legally due date.

Suppose B wants to have the money before the legally due date. Then he can have the money from the banker or a broker, who deducts S.I. on the face value (i.e., Rs. 10,000 in this case) for the period from the date on which the bill was discounted (i.e., paid by the banker) and the legally due date.

This amount is known as **Banker's Discount (B.D.)**.

Thus, B.D. is the S.I. on the face value for the period from the date on which the bill was discounted and the legally due date.

Banker's Gain (B.G.) = (B.D.) - (T.D.) for the unexpired time.

Note: When the date of the bill is not given, grace days are not to be added.

IMPORTANT FORMULAE

1. B.D. = S.I. on bill for unexpired time.
2. $B.G. = (B.D.) - (T.D.) = \text{S.I. on T.D.} = \frac{(T.D.)^2}{P.W.}$
3. $B.D. = \frac{\text{Amount} \times \text{Rate} \times \text{Time}}{100}$
4. $T.D. = \sqrt{P.W. \times B.G.}$
5. $T.D. = \frac{\text{Amount} \times \text{Rate} \times \text{Time}}{100 + (\text{Rate} \times \text{Time})}$
6. $T.D. = \frac{B.D. \times 100}{100 + (\text{Rate} \times \text{Time})}$
7. $TD = \frac{B.G. \times 100}{\text{Rate} \times \text{Time}}$
8. $\text{Amount} = \frac{B.D. \times T.D.}{B.D. - T.D.}$

13. Boat & Stream

1. **Down Stream:** In water, the direction along the stream is called downstream
2. **Up Stream:** In water, the direction against the stream is called upstream
3. If the speed of a boat in still water is u km/hr and the speed of the stream is v km/hr, then:
Speed of boat in downstream = $(u + v)$ km/hr.
Speed of boat in upstream = $(u - v)$ km/hr.
4. If the speed downstream is a km/hr and the speed upstream is b km/hr, then:
Speed in still water = $\frac{a+b}{2}$ km/hr.
Rate of stream = $\frac{a-b}{2}$ km/hr.

14. Surds & Indices

1. $a^m \times a^n = a^{m+n}.$
2. $a^m \div a^n = a^{m-n}.$
3. $(a^m)^n = a^{mn}.$
4. $(ab)^n = a^n \times b^n.$
5. $\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}.$
6. $a^0 = 1.$
7. Let a be any rational number and n be any positive integer such that $\sqrt[n]{a} = a^{\frac{1}{n}}$ is irrational.
Then $\sqrt[n]{a} = a^{\frac{1}{n}}$ is called a surd of order n .
8. $\sqrt[n]{a} = a^{\frac{1}{n}}$
9. $\sqrt[n]{ab} = \sqrt[n]{a} \times \sqrt[n]{b}.$
10. $\sqrt[n]{\frac{a}{b}} = \frac{\sqrt[n]{a}}{\sqrt[n]{b}}.$
11. $(\sqrt[n]{a})^n = a.$
12. $(\sqrt[n]{a})^m = \left(a^{\frac{1}{n}}\right)^m = a^{\frac{m}{n}}.$

15. HCF & LCM

1. **Factors and Multiples:** If number a divided another number b exactly, we say that a is a factor of b .
In this case, b is called a multiple of a .
2. **Highest Common Factor (H.C.F.) or Greatest Common Measure (G.C.M.) or Greatest Common Divisor (G.C.D.):** The H.C.F. of two or more than two numbers is the greatest number that divided each of them exactly.
There are two methods of finding the H.C.F. of a given set of numbers:
 - (a) Factorization Method: Express the each one of the given numbers as the product of prime factors. The product of least powers of common prime factors gives H.C.F.
 - (b) Division Method: Suppose we have to find the H.C.F. of two given numbers, divide the larger by the smaller one. Now, divide the divisor by the remainder. Repeat the process of dividing the preceding number by the remainder last obtained till zero is obtained as remainder. The last divisor is required H.C.F.
 - (c) Finding the H.C.F. of more than two numbers: Suppose we have to find the H.C.F. of three numbers, then, H.C.F. of [(H.C.F. of any two) and (the third number)] gives the H.C.F. of three given number.
Similarly, the H.C.F. of more than three numbers may be obtained.
3. **Least Common Multiple (L.C.M.):** The least number which is exactly divisible by each one of the given numbers is called their L.C.M.
There are two methods of finding the L.C.M. of a given set of numbers:

- (a) Factorization Method: Resolve each one of the given numbers into a product of prime factors. Then, L.C.M. is the product of highest powers of all the factors.
- (b) Division Method (short-cut): Arrange the given numbers in a row in any order. Divide by a number which divided exactly at least two of the given numbers and carry forward the numbers which are not divisible. Repeat the above process till no two of the numbers are divisible by the same number except 1. The product of the divisors and the undivided numbers is the required L.C.M. of the given numbers.
4. Product of two numbers = Product of their H.C.F. and L.C.M.
5. Co-primes: Two numbers are said to be co-primes if their H.C.F. is 1.
6. H.C.F. and L.C.M. of Fractions:
 - (a) $HCF = \frac{HCF \text{ of Numerator}}{LCM \text{ of Denominator}}$
 - (b) $LCM = \frac{LCM \text{ of Numerator}}{HCF \text{ of Denominator}}$
7. **H.C.F. and L.C.M. of Decimal Fractions:**
In a given numbers, make the same number of decimal places by annexing zeros in some numbers, if necessary. Considering these numbers without decimal point, find H.C.F. or L.C.M. as the case may be. Now, in the result, mark off as many decimal places as are there in each of the given numbers.
8. **Comparison of Fractions:** Find the L.C.M. of the denominators of the given fractions. Convert each of the fractions into an equivalent fraction with L.C.M as the denominator, by multiplying both the numerator and denominator by the same number. The resultant fraction with the greatest numerator is the greatest.

16. Identities**Formulae:**

1. $(a+b)^2 = a^2 + 2ab + b^2 = (a-b)^2 + 4ab$
2. $(a-b)^2 = a^2 - 2ab + b^2 = (a+b)^2 - 4ab$
3. $(a+b)^2 + (a-b)^2 = 2(a^2 + b^2)$
4. $(a+b)^2 - (a-b)^2 = 4ab$
5. $ab = \left(\frac{a+b}{2}\right)^2 - \left(\frac{a-b}{2}\right)^2$
6. $a^2 + b^2 = (a+b)^2 - 2ab = (a-b)^2 + 2ab$
7. $a^2 - b^2 = (a+b)(a-b)$
8. $(x+a)(x+b) = x^2 + x(a+b) + ab$
9. $(a+b+c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$
10. $a^2 + b^2 + c^2 = (a+b+c)^2 - 2(ab+bc+ca)$
11. $2(ab+bc+ca) = (a+b+c)^2 - (a^2 + b^2 + c^2)$
12. $(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3 = a^3 + b^3 + 3ab(a+b)$
13. $(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3 = a^3 - b^3 - 3ab(a-b)$

14. $a^3 + b^3 = (a + b)^3 - 3ab(a + b)$
15. $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$
16. $a^3 - b^3 = (a - b)^3 + 3ab(a - b)$
17. $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$
18. $(a + b)^3 + (a - b)^3 = 2a^3 + 6ab^2 = 2a(a^2 + 3b^2)$
19. $(a + b)^3 - (a - b)^3 = 6a^2b + 2b^3 = 2b(3a^2 + b^2)$
20. $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$
21. $a^3 + b^3 + c^3 - 3abc = \frac{1}{2}(a + b + c)\{(a - b)^2 + (b - c)^2 + (c - a)^2\}$
22. If $a + b + c = 0$, then $a^3 + b^3 + c^3 = 3abc$.
23. $a^4 + a^2b^2 + b^4 = (a^2 + ab + b^2)(a^2 - ab + b^2)$
24. $a^4 - b^4 = (a^2 + b^2)(a + b)(a - b)$
25. $a^n - 1 = (a - 1)(1 + a + a^2 + \dots + a^{n-1})$
26. $a^n - b^n = (a - b)(a^{n-1} + a^{n-2}b + a^{n-3}b^2 + \dots + b^{n-1})$

17. Number Theory

- (i) Natural Number = $N = 1, 2, 3, 4, 5, \dots$
- (ii) Whole Number = $N^* = 0, 1, 2, 3, 4, \dots$
- (iii) Integers = $Z = \dots, -3, -2, -1, 0, 1, 2, 3, \dots$
- (iv) Odd Numbers: A number which is not divisible by 2 is called as odd numbers. Example: 1, 3, 5, 7, ..., $(2n+1), \dots$ n is an integer.
- (v) Even Numbers: A number which is divisible by 2 is called as even numbers. Example: 2, 4, 6, 8, ..., $2n, \dots$ n is an integer.

(vi) Properties of Odd and Even Numbers:

- (i) Sum of two even numbers is even.
- (ii) Sum of two odd numbers is even.
- (iii) Sum of one odd and one even number is odd.
- (iv) Product of two odd numbers is odd.
- (v) Product of two even numbers is even.
- (vi) Product of one odd and one even number is even.
- (vii) If a is odd, then $a^2, a^3, a^5, \dots$ are odd numbers.
- (viii) If a is even, then $a^2, a^3, a^5, \dots$ are even numbers.

(vii) Different Types of Brackets:

‘ _____ ’ – Virnaculum or bar
 () – Paranthesis
 [] – Square Brackets
 { } – Braces

(viii) VBODMAS Rule:

V – Virnaculum
 B – Brackets
 O – Of or orders of numbers like square, square root of numbers
 D – Division
 M – Multiplication
 A – Addition
 S – Subtraction

(ix) Divisibility Test:

1. A positive integer N is divisible by 2 if the unit digit in N is 0, 2, 4, 6, or 8.
2. A positive integer N is divisible by 3 if the sum of all the digits in N is divisible by 3.
3. A positive integer N is divisible by 4 if the last two digits of N is divisible by 4.
4. A positive integer N is divisible by 5 if the unit digit in N is 0 or 5.
5. A positive integer N is divisible by 6 if it is divisible by 2 and 3 both.
6. A positive integer N is divisible by 8 if the last three digits of N is divisible by 8.
7. A positive integer N is divisible by 9 if the sum of all the digits in N is divisible by 9.
8. A positive integer N is divisible by 10 if the unit digit in N is 0.
9. A positive integer N is divisible by 11 if the difference between the sum of digits in the odd places and the sum of digits in the even places is divisible by 11.
10. A positive integer N is divisible by 12 if it is divisible by 3 and 4 both.
11. A positive integer N is divisible by 15 if it is divisible by 3 and 5 both.
12. A positive integer N is divisible by 16 if the last four digits of N is divisible by 16.
13. A positive integer N is divisible by 18 if it is divisible by 2 and 9 both.
14. A positive integer N is divisible by 20 if it is divisible by 4 and 5 both.....
15. If any digits repeated thrice, then it is divisible by 3 as well as 37. Example: 111, 121212, 222, 444, 767676, 345345345, etc are disable by 3 as well as 37.
16. If a and b are positive integers then $a^n - b^n$ is always divisible by $a - b$.
17. $a | 0$, zero is divisible by all non-zero integers.
18. $1 | a$, any integer is always divisible by 1.
19. $a | a$, any non-zero integer is always divisible by itself.
20. If $a | b$ and $b | c$, then $a | c$.
21. If $a | b$ and $a | c$, then $a | (mb + nc)$, where m and n are integers.
22. If $ca | cb$, then $a | b$, where $c \neq 0$.
23. If $a | b$, then $ca | cb$.
24. Every Composite integer can be expressed as product of primes. i.e.

$$N = p_1^{n_1} \times p_2^{n_2} \times p_3^{n_3} \times \dots \times p_m^{n_m}$$
, where $N = p_1, p_2, p_3, \dots, p_m$, are primes.

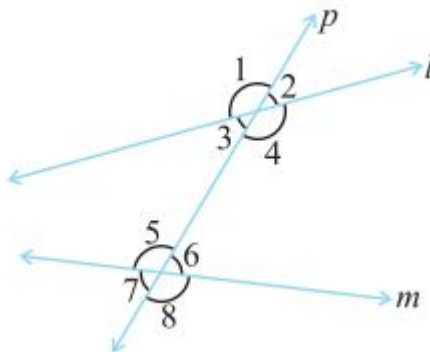
25. **Euclid's Division Lemma**: Given positive integers a and b there exist unique integer q and r such that $a = bq + r$, where $0 \leq r < b$.
26. **Euclid's Algorithm**: It is used to find the HCF of two numbers. Let it be a and b . Let $a > b$
- By Euclid's lemma $a = bq + r$, where $0 \leq r < b$.
 - If $r = 0$, then HCF of a and b is q .
 - If $r \neq 0$, then apply Euclid's lemma to b and r .
 - Continue till the HCF obtained.
27. If $N = p_1^{n_1} \times p_2^{n_2} \times p_3^{n_3} \times \dots \times p_m^{n_m}$ is any number, then number of divisor of N are $(n_1 + 1)(n_2 + 1)(n_3 + 1) \dots (n_m + 1)$.
28. Number of ways in which $N = p_1^{n_1} \times p_2^{n_2} \times p_3^{n_3} \times \dots \times p_m^{n_m}$ can be resolved into product of two factors $= \frac{1}{2}(n_1 + 1)(n_2 + 1)(n_3 + 1) \dots (n_m + 1)$.
29. Number of ways in which $N = p_1^{n_1} \times p_2^{n_2} \times p_3^{n_3} \times \dots \times p_m^{n_m}$ can be resolved into product of two factors which are prime to each other $= 2^{m-1}$. (where there are m numbers of distinct primes in N).
30. Sum of divisors of a number $N = p_1^{n_1} \times p_2^{n_2} \times p_3^{n_3} \times \dots \times p_m^{n_m}$ is $\frac{p_1^{n_1+1}-1}{p_1-1} \times \frac{p_2^{n_2+1}-1}{p_2-1} \times \frac{p_3^{n_3+1}-1}{p_3-1} \times \dots \times \frac{p_m^{n_m+1}-1}{p_m-1}$.
31. The highest power of prime p which is divisible by $N!$ is $\left[\frac{N}{p} \right] + \left[\frac{N}{p^2} \right] + \left[\frac{N}{p^3} \right] + \dots$
32. The product of N consecutive integers is always divisible by $N!$.
33. **Fermat's Little Theorem**: If p is a prime, and n be prime to p , then $n^{p-1} - 1$ is divisible by p .
34. If p is a prime and n is any integer, then $n^p - n$ is divisible by p .
35. **Wilson Theorem**: If p is a prime, then $(p-1)! + 1$ is divisible by p .
36. **Euler's Totient Function $\Phi(n)$** : If $n > 1$ be any positive integer, then the number of all positive integers less than n and prime to n is denoted by $\Phi(n)$. This $\Phi(n)$ is called as Euler's Totient function.
- Properties of Euler's Totient Function $\Phi(n)$**
- If a and b are prime to each other then $\Phi(ab) = \Phi(a)\Phi(b)$.
 - If $n = p_1^{n_1} \times p_2^{n_2} \times p_3^{n_3} \times \dots \times p_m^{n_m}$, where $p_1, p_2, p_3, \dots, p_m$, are distinct primes, then $\Phi(n) = n \times \left(1 - \frac{1}{p_1}\right) \left(1 - \frac{1}{p_2}\right) \left(1 - \frac{1}{p_3}\right) \dots$
 - If n is a positive integer prime to N , then $n^{\Phi(N)} - 1$ is divisible by N .
 - If n is a prime number, then $\Phi(n) = n - 1$.
37. **Lagrange Theorem**: Let $p > 3$ be a prime and let $(x-1)(x-2) \dots (x-p+1) = x^{p-1} - A_1 x^{p-2} - A_2 x^{p-3} - \dots - A_{p-2} x + A_{p-1}$ then $A_{p-1} + 1$ is divisible by p and A_{p-2} is divisible by p^2 .

- (x) **Prime Numbers:** If a number has only two factors 1 and the number itself, then it is called as a prime number. Example: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, ...
- (xi) **Composite Numbers:** If a number has more than two factors, then it is called as composite numbers. Example: 4, 6, 8, 9, 10, 12, 14, 15, 16, 18, 20,
- (xii) **Perfect Number:** A number is said to be a perfect number, if sum of its divisors is the twice of the given number.
Example: 6 has factor 1, 2, 3, 6. Sum of the factors = $1+2+3+6 = 12 = 2 \times 6$.
28 has factor 1, 2, 4, 7, 14, 28. Sum of factors = $1+2+4+7+14+28 = 56 = 2 \times 28$.
- (xiii) **Pythagorean Triplet:** Three numbers (a, b, c) are called as Pythagorean triplet if $a^2 + b^2 = c^2$.
- (xiv) **Fibonacci Number:** $F_{n+1} = F_n + F_{n-1}$ ($n > 2$), $F_1 = F_2 = 1$.
Fibonacci numbers are 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, 233, 377, ...
- (xv) **Relatively Prime:** Two numbers are said to be relatively prime to each other if their GCD of HCF is 1. Example: 4, 5 has HCF 1. So they are relatively prime.
- (xvi) Dividend = Divisor X Quotient + Remainder.
- (xvii) Remainder = Dividend - Divisor X Quotient.
- (xviii) Quotient = (Dividend – Remainder) / Divisor.
- (xix) Divisor = (Dividend – Remainder) / Quotient.
- (xx) **Pigeonhole Principle:** If $n+1$ objects are distributed at random into n boxes, then atleast one box has atleast two objects.

18. Geometry**1. Angles:**

- (a) Two rays OA and OB with common point O form an angle AOB denoted as $\angle AOB$.
- (b) Angle can be measured by degrees, minute and second. $1^\circ = 60'$, $1' = 60''$.
- (c) Right Angle: An angle whose measurement is 90° is called as right angle.
- (d) Acute angle: An angle whose measurement is greater than 0° and less than 90° is called as acute angle.
- (e) Obtuse angle: An angle whose measurement is greater than 90° and less than 180° is called as obtuse angle.
- (f) Straight angle: An angle whose measurement is 180° is called as straight angle.
- (g) Reflex angle: An angle whose measurement is greater than 180° and less than 360° is called as reflex angle.
- (h) Complete angle: An angle whose measurement is 360° is called as complete angle.
- (i) Equal Angles: Two angles are said to be equal if they have equal measurement.
- (j) Complementary angles: Two angles are said to be complementary if their sum is 90° .
- (k) Supplementary angles: Two angles are said to be supplementary if their sum is 180° .
- (l) Adjacent angles: Two angles are said to be adjacent if they have one common arm and one common vertex.
- (m) If a ray stands on a straight line, then sum of two adjacent angles is 180° .
- (n) The sum of all angles around a point is 360° .
- (o) **Vertically opposite angles:** If two lines intersect at a point, then two pairs of vertically opposite angles so formed are equal to each other.
- (p) **Parallel Lines:** If two lines on the same plane do not intersect when produced either side are called as parallel lines.

(q) **Angles Formed when a transversal line intersect two lines:**



Interior angles	$\angle 3, \angle 4, \angle 5, \angle 6$
Exterior angles	$\angle 1, \angle 2, \angle 7, \angle 8$
Pairs of Corresponding angles	$\angle 1$ and $\angle 5$, $\angle 2$ and $\angle 6$, $\angle 3$ and $\angle 7$, $\angle 4$ and $\angle 8$
Pairs of Alternate interior angles	$\angle 3$ and $\angle 6$, $\angle 4$ and $\angle 5$
Pairs of Alternate exterior angles	$\angle 1$ and $\angle 8$, $\angle 2$ and $\angle 7$
Pairs of interior angles on the same side of the transversal	$\angle 3$ and $\angle 5$, $\angle 4$ and $\angle 6$

(r) **When a transversal line intersect two parallel lines:**

Pairs of Alternate Interior angles are equal.

Pairs of Alternate Exterior angles are equal.

Pairs of corresponding angles are equal.

Pairs of interior angles on the same side of the transversal are supplementary.

Pairs of exterior angles on the same side of the transversal are supplementary.

2. **Incentre:** The point of concurrency of angle bisectors of a triangle is called as Incentre.
3. **Circumcentre:** The point of concurrency of perpendicular bisectors sides of a triangle is called as Circumcentre.
4. **Orthocentre:** The point of concurrency of altitudes of a triangle is called as Orthocentre.
5. **Centroid:** The point of concurrency of medians of a triangle is called as Centroid.
6. **Congruence Triangles:** Two triangles are congruent if their corresponding sides and angles are equal.
7. **SSS Congruent Condition:** Two triangles are congruent if three sides of one triangle are respectively equal to the three sides of the other triangle.
8. **SAS Congruent Condition:** Two triangles are congruent if two sides and included angle of one triangle are respectively equal to the two sides and included angle of the other triangle.
9. **ASA Congruent Condition:** Two triangles are congruent if two angles and included side of one triangle are respectively equal to the two angles and included side of the other triangle.
10. **RHS Congruent Condition:** Two right triangles are congruent if the hypotenuse and one of the side of one triangle are respectively equal to the three hypotenuse and one of the side of the other triangle.

11. Properties of Triangle:

- (a) Sum of all the interior angles of a triangle is 180° .
- (b) The longest side of a triangle is opposite to the greatest angle and vice versa. The least side of triangle is opposite to the smaller angle and vice versa.
- (c) Sum of any two sides of a triangle is always greater than the third side.
- (d) If one angle of a triangle is greater than the second angle then the side opposite to the first is greater than the side opposite to second.
- (e) Exterior angle of a triangle is equal to the sum of two interior opposite angles.
- (f) A scalene triangle is one which has all sides are unequal.
- (g) An isosceles triangle has two of its sides are equal.
- (h) An equilateral triangle has all the three sides are equal.
- (i) An obtuse angled triangle has one of its angle is obtuse.
- (j) An acute angled triangle has all of its angles are acute.
- (k) A right angled triangle has one right angle.
- (l) There can be only one right angle in a triangle.
- (m) There can be only one obtuse angle in a triangle.
- (n) Angles opposite to equal sides of a triangle are equal.
- (o) Sides opposite to equal angles of a triangle are equal.
- (p) The line segment joining the mid points of two sides of a triangle is parallel to third and half of third side.
- (q) A line drawn through the midpoint of two sides of a triangle is parallel to the third side and half of third side.

12. Properties of Parallelogram:

- (a) Opposite sides of a parallelogram are equal.
- (b) Opposite angles of a parallelogram are equal.
- (c) A pair of adjacent angles of a parallelogram is supplementary.
- (d) A diagonal divided the parallelogram into two congruent triangles.
- (e) Diagonals of a parallelogram bisect each other.

13. Properties of Rectangle:

- (a) Opposite sides of a rectangle are equal.
- (b) Each angles of a rectangle is right angle.
- (c) Diagonals of a rectangle bisect each other.
- (d) Diagonals of a rectangle are equal.

14. Properties of a square:

- (a) All the angles of a square are right angles.
- (b) All the sides of a square are equal.
- (c) Diagonals of a square are equal.
- (d) Diagonal bisect each other.
- (e) Diagonals of a square are perpendicular to each other.

15. Properties of Rhombus:

- (a) All the sides of the rhombus are equal.
- (b) Opposite angles of the rhombus are equal.
- (c) Diagonal of a rhombus bisect each other.
- (d) Diagonals of a rhombus are perpendicular to each other.
- (e) Diagonal bisects the angles at the vertices.

16. Properties of Parallel Lines:

- (a) Two lines in a plane which are parallel to the given line is parallel to each other.
- (b) Two intersecting lines cannot both parallel to the same line.
- (c) Two lines in the same plane, which are perpendicular to a given line in the same plane, are parallel to each other.
- (d) Three or more parallel lines makes equal intercepts on any transversal then they make equal intercepts on any other transversal also.
- (e) If a line parallel to one of the side of a triangle intersects other two sides in the two different points then intercepts proportional to these two sides.
- (f) If three or more parallel lines intersects two transversal, then the intercepts made on one transversal are proportional to the intercepts made on other transversal.

17. Properties of a Circle:

- (a) Perpendicular from the centre of a circle to any chord, bisect the chord.
- (b) The line segment joining the centre of a circle to the midpoint of the chord is perpendicular to the chord.
- (c) Equal chords of a circle subtend equal angles at the centre.
- (d) In a circle, chords which subtend equal angles at centre are equal.
- (e) Equal chords of a circle are equidistant from the centre.
- (f) Chords which are equidistant from centre are equal.
- (g) An angle subtended by an arc at the centre is twice the angle subtended by it at any point on the remaining part of the circle.
- (h) If two chords subtend equal angles at the circumference of a circle, the chords are equal.
- (i) The line perpendicular to the tangent at the point of contact passes through the centre of the circle.
- (j) The line drawn perpendicular to the radius through the end point where the radius meets the circle is tangent to the circle at the end point.
- (k) The lengths of the two tangents drawn from an external point to the circle are equal.
- (l) If from an external point P to a circle PAB is a secant line and PT is a tangent to the circle then $PA \times PB = PT^2$.

19. Area Surface & Volume**Triangle:**

- Perimeter of a triangle = Sum of all the three sides.
- Area of a triangle = $\frac{1}{2} \times \text{Base} \times \text{Height}$.
- Area of a triangle = $\sqrt{s(s-a)(s-b)(s-c)}$, where $s = \frac{a+b+c}{2}$.
- Perimeter of equilateral triangle = $3 \times \text{side}$.
- Area of equilateral triangle = $\frac{\sqrt{3}}{4} \times (\text{side})^2 = \frac{1}{\sqrt{3}} \times (\text{height})^2$.
- Height of equilateral triangle = $\frac{\sqrt{3}}{2} \times \text{side}$.
- Side of equilateral triangle = $\frac{2}{\sqrt{3}} \times \text{Height}$.
- Area of isosceles triangle = $\frac{1}{2} \times \text{Base} \times \sqrt{(\text{equal side})^2 - \frac{(\text{Base})^2}{4}}$.
- Height of isosceles triangle = $\sqrt{(\text{equal side})^2 - \frac{(\text{Base})^2}{4}}$.
- Base of isosceles triangle = $2\sqrt{(\text{equal side})^2 - (\text{height})^2}$.
- Equal side isosceles triangle = $\sqrt{(\text{height})^2 + \frac{(\text{Base})^2}{4}}$.
- Hypotenuse of right angled isosceles triangle = $\sqrt{2} \times \text{side}$.
- Equal Side of right angled isosceles triangle = $\frac{\text{Hypotenuse}}{\sqrt{2}}$.
- Area of right angled isosceles Triangle = $\frac{1}{2} \times (\text{equal side})^2 = \frac{1}{4} \times (\text{Hypotenuse})^2$

Square:

- Perimeter of a square = $4 \times \text{side}$
- Area of square = $(\text{side})^2$.
- Diagonal of a square = $\sqrt{2} \times \text{side}$.

Rectangle:

- Perimeter of rectangle = $2 \times (\text{length} + \text{breadth})$.
- Area of rectangle = $\text{length} \times \text{breadth}$.
- Diagonal of rectangle = $\sqrt{(\text{length})^2 + (\text{breadth})^2}$.
- Area of Four Walls = $2 \times \text{height} \times (\text{length} + \text{breadth})$.

Parallelogram:

- Area of a parallelogram = $\text{Base} \times \text{Height}$.

Rhombus:

- Area of rhombus = $\frac{1}{2} \times \text{Product of diagonals}$.
- Side of a rhombus = $\frac{1}{2} \sqrt{(d_1)^2 + (d_2)^2}$, where d_1, d_2 are lengths of diagonals.

Trapezium:

- Area of trapezium = $\frac{1}{2} \times \text{Sum of parallel sides} \times \text{Height}$

Circle:

- Circumference of a circle = $2\pi.r = \pi.d$.
- Area of circle = $\pi.r^2 = \frac{\pi.d^2}{4}$.
- Radius of circle = $\frac{\text{diameter}}{2}$.

Arc:

- Length of arc = $\frac{\theta}{360^\circ} \times 2\pi.r$.
- Area of arc = $\frac{\theta}{360^\circ} \times \pi.r^2$

Cube: Let side of the cube = a .

- Surface area of cube = $6a^2$.
- Area of four walls of cube = $4a^2$.
- Volume of the cube = a^3 .
- Main diagonal of a cube = $\sqrt{3} \times a$.

Cuboid: Let length = l , breadth = b and height = h

- Total surface area of cuboid = $2(lb + bh + hl)$
- Area of four walls of cuboid = $2h(l + b)$.
- Volume of the cuboid = $l \times b \times h$.
- Main diagonal of a cuboid = $\sqrt{l^2 + b^2 + h^2}$.

Right Circular Cylinder: Let radius of base = r and height = h

- Base area of cylinder = $\pi.r^2$
- Curved surface area of cylinder = $2\pi.rh$
- Total surface area of cylinder = $2\pi.rh + 2\pi.r^2 = 2\pi.r(r + h)$.
- Volume of cylinder = $\pi.r^2h$

Right Circular Cone: Let radius of base = r , height = h and slant height = l

- Slant height = $l = \sqrt{r^2 + h^2}$
- Base area of cone = $\pi.r^2$
- Curved surface area of cone = $\pi.rl$
- Total Surface area of cone = $\pi.r^2 + \pi.rl = \pi.r(r + l)$.
- Volume of the cone = $\frac{1}{3}\pi.r^2h$

Sphere:

- Curved surface area of sphere = $4\pi.r^2$
- Volume of the sphere = $\frac{4}{3}\pi.r^3$.

Hemisphere:

- Curved surface area of hemisphere = $2\pi.r^2$
- Base area of hemisphere = $\pi.r^2$
- Total surface area of hemisphere = $3\pi.r^2$
- Volume of the hemisphere = $\frac{2}{3}\pi.r^3$.

Frustum: Let h be height, l be the slant height and R and r are the radii of both the ends of a frustum.

- Slant height of frustum = $l = \sqrt{h^2 + (R - r)^2}$
- Smaller base area of frustum = $\pi.r^2$
- Bigger base area of frustum = $\pi.R^2$
- Curved surface area of frustum = $\pi.(R + r)l$.
- Total surface area of frustum = $\pi.r^2 + \pi.R^2 + \pi.(R + r)l = \pi.(r^2 + R^2 + Rl + rl)$.
- Volume of frustum = $\frac{1}{3}\pi.h(R^2 + r^2 + Rr)$.

n-Gon: Let a n -gon is there.

- Sum of interior angles of n -gon = $(n - 2) \times 180^\circ$.
- Sum of Exterior angles of n -gon = 360°
- For a regular n -gon each interior angles = $\frac{(n - 2)}{n} \times 180^\circ$.
- For a regular n -gon each exterior angles = $\frac{360^\circ}{n}$.

Arithmetic Progression

- **Arithmetic Progression(AP)**: A sequence of numbers is said to be in Arithmetic progression (AP) if the difference between two consecutive terms are same.
Arithmetic progression is of the form: $a, a + d, a + 2d, a + 3d, \dots$
- **nth Term of an Arithmetic Progression**: nth term of an arithmetic progression can be written as $t_n = a + (n-1)d$, where a is the first term and d is the common difference of an arithmetic progression.
 - (1) If an AP has n terms then the last term of AP is $l = a + (n-1)d$.
 - (2) If three numbers a, b, c are in AP then $b - a = c - b \Rightarrow a + c = 2b$.
 - (3) If a is the first term and d is the common difference of an arithmetic progression having m terms, then the n th term from end is the $(m - n + 1)$ th term.
So, n th term from end is $a + (m - n)d$.
 - (4) Three number in AP $a - d, a, a + d$.
 - (5) Four number in AP $a - 3d, a - d, a + d, a + 3d$.
 - (6) Five number in AP $a - 3d, a - d, a, a + d, a + 3d$.
- **Sum of n terms of an Arithmetic Progression**:
 - (1) $S_n = \frac{n}{2} \{2a + (n-1)d\}$.
 - (2) $S_n = \frac{n}{2} \{a + l\}$.
 - (3) $t_n = S_n - S_{n-1}$.
- **Properties of Arithmetic Progression**:
 - (a) If $a_1, a_2, a_3, \dots, a_n$ are in AP, then:
 - (1) $a_1 + k, a_2 + k, a_3 + k, \dots, a_n + k$ are in AP.
 - (2) $a_1 - k, a_2 - k, a_3 - k, \dots, a_n - k$ are in AP.
 - (3) $a_1 \times k, a_2 \times k, a_3 \times k, \dots, a_n \times k$ are in AP.
 - (4) $a_1 / k, a_2 / k, a_3 / k, \dots, a_n / k$ are in AP.
 - (5) $a_1 + a_n = a_2 + a_{n-1} = a_3 + a_{n-2} = \dots$
 - (6) $a_r = \frac{a_{r-k} + a_{r+k}}{2}$, where $0 \leq k \leq n - r$.
 - (7) If the number of terms of any AP is odd, then sum of the terms is equal to product of middle terms and number of terms.
 - (8) If the number of terms of any AP is odd then its middle term is A.M. (arithmetic mean) of first and last term.
 - (b) If $a_1, a_2, a_3, \dots, a_n$ and $b_1, b_2, b_3, \dots, b_n$ are two AP's, then:
 - (1) $a_1 + b_1, a_2 + b_2, a_3 + b_3, \dots, a_n + b_n$ are in AP.
 - (2) $a_1 - b_1, a_2 - b_2, a_3 - b_3, \dots, a_n - b_n$ are in AP.

- **Arithmetic Mean:**

- (1) **Single Arithmetic Mean:** If A is a single arithmetic mean between two numbers a, b then a, A, b are in AP.
- (2) **n Arithmetic Mean:** The numbers $A_1, A_2, A_3, \dots, A_n$ are n arithmetic mean between two numbers a, b , then $a, A_1, A_2, A_3, \dots, A_n, b$ are in AP.
- (3) **Inserting Single A.M. between two given numbers:** If A is the arithmetic mean between two numbers a, b then $A = \frac{a+b}{2}$.
- (4) **Inserting n A.M. between two given numbers:** If $A_1, A_2, A_3, \dots, A_n$ are n arithmetic mean between two numbers a, b then $d = \frac{b-a}{n+1}$.
 So, $A_1 = a + \frac{b-a}{n+1}$, $A_2 = a + 2\left(\frac{b-a}{n+1}\right)$, $A_3 = a + 3\left(\frac{b-a}{n+1}\right)$, ..., $A_n = a + n\left(\frac{b-a}{n+1}\right)$
 Sum of n arithmetic mean $A_1 + A_2 + A_3 + \dots + A_n = n\left(\frac{a+b}{2}\right)$.

- **Some Special Sequence:**

- (1) $1 + 2 + 3 + \dots + n = \sum_{k=1}^n k = \frac{n(n+1)}{2}$.
- (2) $1^2 + 2^2 + 3^2 + \dots + n^2 = \sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$.
- (3) $1^3 + 2^3 + 3^3 + \dots + n^3 = \sum_{k=1}^n k^3 = \left(\frac{n(n+1)}{2}\right)^2$.
- (4) $1^4 + 2^4 + 3^4 + \dots + n^4 = \sum_{k=1}^n k^4 = \frac{n(n+1)(2n+1)(3n^2+3n-1)}{30}$.

Ratio Proportion

- For the ratio $a:b$, a – Antecedents, b – Consequents.
- If b is the mean proportional between a and c then $b^2 = ac$.
- If a, b, c, d are in proportion then $\frac{a}{b} = \frac{c}{d}$.
- If a, b, c are in proportion then $\frac{a}{b} = \frac{b}{c}$.
- If $\frac{a}{b} > 1$ then $\frac{a}{b} > \frac{a+x}{b+x}$, ($x > 0$).
- If $\frac{a}{b} < 1$ then $\frac{a}{b} < \frac{a+x}{b+x}$, ($x > 0$).

- If $\frac{a_1}{b_1}, \frac{a_2}{b_2}, \frac{a_3}{b_3}, \dots, \frac{a_n}{b_n}$ are unequal fractions, of which the denominators are of the same sign, then the fraction $\frac{a_1 + a_2 + a_3 + \dots + a_n}{b_1 + b_2 + b_3 + \dots + b_n}$ lies in magnitude between the greatest and least of them.
- If $\frac{a_1}{b_1} = \frac{a_2}{b_2} = \frac{a_3}{b_3} = \dots = \frac{a_n}{b_n}$, then each of these ratios equals to $\frac{a_1}{b_1} = \frac{a_2}{b_2} = \frac{a_3}{b_3} = \dots = \frac{a_n}{b_n} = \left(\frac{\alpha_1 a_1^n + \alpha_2 a_2^n + \alpha_3 a_3^n + \dots + \alpha_n a_n^n}{\alpha_1 b_1^n + \alpha_2 b_2^n + \alpha_3 b_3^n + \dots + \alpha_n b_n^n} \right)^{1/n}$, where n is an integer.
- If a, b, c, d are in proportion. Then,

$$(1) \quad \frac{a}{b} = \frac{c}{d}.$$

$$(2) \quad \frac{a}{c} = \frac{b}{d}. \quad (\text{Alternendo})$$

$$(3) \quad \frac{a+b}{b} = \frac{c+d}{d}. \quad (\text{Componendo})$$

$$(4) \quad \frac{a-b}{b} = \frac{c-d}{d}. \quad (\text{Dividendo})$$

$$(5) \quad \frac{a+b}{a-b} = \frac{c+d}{c-d}. \quad (\text{Componendo \& Dividendo})$$

$$(6) \quad \frac{b}{a} = \frac{d}{c}. \quad (\text{Invertendo})$$

- Comparison of Ratios: We say $(a:b) > (c:d)$ means $\frac{a}{b} > \frac{c}{d}$
- Compounded Ratio: The compounded ratio of the ratios: $(a : b), (c : d), (e : f)$ is $(ace : bdf)$.
- Duplicate Ratios: Duplicate ratio of $(a : b)$ is $a^2 : b^2$.
- Sub-duplicate ratio of $(a : b)$ is $a^{1/2} : b^{1/2}$.
- Triplicate ratio of $(a : b)$ is $a^3 : b^3$.
- Sub-triplicate ratio of $(a : b)$ is $a^{1/3} : b^{1/3}$.
- **Factorial Notation:** The product of first n natural numbers is called as factorial n and is denoted by $n! = \underline{n} = 1 \times 2 \times 3 \times 4 \times \dots \times n$.
 - (1) We have to assume $0! = 1$, though it makes no sense.
 - (2) When n is a negative integer then $n! = \underline{n}$ is not defined.
 - (3) $n! = \underline{n} = 1 \times 2 \times 3 \times 4 \times \dots \times n = n \times 1 \times 2 \times 3 \times 4 \times \dots \times (n-1)$
 $\Rightarrow n! = n \times (n-1)!$
 - (4) $(2n)! = 2^n \times n! \times (1 \times 3 \times 5 \times \dots \times (2n-1))$.

EASY WAY TO FIND UNIT PLACE OF m^n WHERE m, n ARE NATURAL NUMBERS

Let m^n is the given number, where m, n are natural numbers. Here we can find the unit place of m^n by a simple method.

Process: -

Unit place of $m^n = \text{Unit place of (Unit place of } m)^n = \text{Unit place of } (a^n)$

Where 'a' is unit place of m and $a \in \{0,1,2,3,4,5,6,7,8,9\}$

1. If $a = 0$, then

Unit place of $(a^n) = 0$

2. If $a = 1$, then

Unit place of $(a^n) = 1$

3. If $a = 2$, then

$$\text{Unit place of } (a^n) = \begin{cases} 2 & \text{if } n = 4k + 1 \\ 4 & \text{if } n = 4k + 2 \\ 8 & \text{if } n = 4k + 3 \\ 6 & \text{if } n = 4k \end{cases}$$

Here k is a natural number.

4. If $a = 3$, then

$$\text{Unit place of } (a^n) = \begin{cases} 3 & \text{if } n = 4k + 1 \\ 9 & \text{if } n = 4k + 2 \\ 7 & \text{if } n = 4k + 3 \\ 1 & \text{if } n = 4k \end{cases}$$

Here k is a natural number.

5. If $a = 4$, then

$$\text{Unit place of } (a^n) = \begin{cases} 4 & \text{if } n = 2k + 1 \\ 6 & \text{if } n = 2k \end{cases}$$

Here k is a natural number.

6. If $a = 5$, then

Unit place of $(a^n) = 5$

7. If $a = 6$, then

Unit place of $(a^n) = 6$

8. If $a = 7$, then

$$\text{Unit place of } (a^n) = \begin{cases} 7 & \text{if } n = 4k + 1 \\ 9 & \text{if } n = 4k + 2 \\ 3 & \text{if } n = 4k + 3 \\ 1 & \text{if } n = 4k \end{cases}$$

Here k is a natural number.

9. If $a = 8$, then

$$\text{Unit place of } (a^n) = \begin{cases} 8 & \text{if } n = 4k + 1 \\ 4 & \text{if } n = 4k + 2 \\ 2 & \text{if } n = 4k + 3 \\ 6 & \text{if } n = 4k \end{cases}$$

Here k is a natural number.

10. If $a = 9$, then

$$\text{Unit place of } (a^n) = \begin{cases} 9 & \text{if } n = 2k + 1 \\ 1 & \text{if } n = 2k \end{cases}$$

Here k is a natural number.

Example 1: - Find the unit place in the expansion of 3897654^{876548} .

$$\begin{aligned} \text{Ans: - Unit place of } 3897654^{876548} &= \text{Unit place of (Unit place of } (3897654) \text{)}^{876548} \\ &= \text{Unit place of } (4)^{876548} \\ &= \text{Unit place of } (4)^{2 \times 438274} \\ &= \text{Unit place of } (4)^{2 \times k} \\ &= 6 \end{aligned}$$

Example 2: - Find the unit place in the expansion of 653768957^{76531} .

$$\begin{aligned} \text{Ans: - Unit place of } (653768957)^{76531} &= \text{Unit place of (Unit place of } (653768957) \text{)}^{76531} \\ &= \text{Unit place of } (7)^{76531} \\ &= \text{Unit place of } (7)^{4 \times 19132 + 3} \\ &= \text{Unit place of } (7)^{4 \times k + 3} \\ &= 3 \end{aligned}$$

Example 3: - Find the unit place in the expansion of $689^{745} + 745^{689}$

$$\begin{aligned} \text{Ans: - Unit place of } 689^{745} &= \text{Unit place of (Unit place of } (689) \text{)}^{745} \\ &= \text{Unit place of } (9)^{745} \\ &= \text{Unit place of } (9)^{2 \times 372 + 1} \\ &= \text{Unit place of } (9)^{2 \times k + 1} \\ &= 9 \end{aligned}$$

$$\begin{aligned} \text{Unit place of } 745^{689} &= \text{Unit place of (Unit place of } (745) \text{)}^{689} \\ &= \text{Unit place of } (5)^{689} \\ &= 5 \end{aligned}$$

$$\begin{aligned} \text{Unit place of } 689^{745} + 745^{689} &= \text{Unit place of } 689^{745} + \text{Unit place of } 745^{689} \\ &= \text{Unit Place of } (9+5) \\ &= \text{Unit Place of } 14 \\ &= 4 \text{ (Ans).} \end{aligned}$$

Hence in this way we can calculate unit place of any natural number to the power any natural number.

Exercise:

1. Find unit place of $(243768953)^{145533}$.
2. Find unit place of $(12300457)^{345627}$.
3. Find unit place of $(523432959)^{983456}$.
4. Find unit place of $(223432)^{733406}$.
5. Find unit place of $(329595234)^{456983}$.
6. Find unit place of $(7689511)^{334455}$.
7. Find unit place of $(76895)^{100034}$.
8. Find unit place of $(1024376)^{99806}$.
9. Find unit place of $(872790)^{4550}$.
10. Find unit place of $(663768)^{663768}$.
11. Find unit place of $(243)^{533} - (65)^{342}$.
12. Find unit place of $(243)^{533} + (65)^{342}$.
13. Find unit place of $(243)^{533} \times (65)^{342}$.

CLOCKS



In this topic basic idea is what is angle between minute hand and hour hand at a particular time. In every clock there are three hands namely Hour hand, Minute hand and Second hand.

1. Hour hand describes an angle 360° in 12 hours. So in 1 hour, Hour hand describe an angle of 30° .
2. Minute hand describes an angle 360° in 60 minutes. So in 1 minute, Minute hand describe an angle of 6° .
3. Second hand describes an angle 360° in 60 seconds. So in 1 second, Second hand describe an angle of 6° .
4. At 12 O' Clock Hour hand and Minute hand coincides. Angle between them is 0° .
5. At 6 O' Clock Hour hand and Minute hand are opposite to each other. Angle between them is 180° .
6. At 3 O' Clock and 9 O' Clock Hour hand and Minute hands are perpendicular to each other. Angle between them is 90° .
7. Clock is divided into 60 equal divisions called as minute divisions.
8. In one hour the minute hand moves 60 divisions and hour hand moves 5 divisions. Hence Minute hand gains 55 minutes over the hour hand in 60 minutes or in one hour.

Q. At what time after 5 O' Clock both Hour and Minute hand coincide with each other?

Q. At what time between 5 O' Clock and 6 O' Clock both Hour and Minute hand coincide with each other?

Ans: At 5 O' Clock the minute hand is at 12 and hour hand is at 5.

So, the minute hand is 25 minutes behind the hour hand.

Both Hour and Minute hand coincide with each other if minute hand gains 25 minutes over the hour hand.

As we know Minute hand gains 55 minutes over the hour hand in 60 minutes.

So, Minute hand gains 1 minute over the hour hand in $\frac{60}{55}$ minutes.

Minute hand gains 25 minutes over the hour hand in $\frac{60}{55} \times 25 = \frac{300}{11} = 27\frac{3}{11}$ minutes.

Hence both the hands coincide at 5 hour $27\frac{3}{11}$ minutes.

Q. At what time after 5 O' Clock both Hour and Minute hand are perpendicular to each other?

Q. At what time between 5 O' Clock and 6 O' Clock both Hour and Minute hand are perpendicular to each other?

Ans: At 5 O' Clock the minute hand is at 12 and hour hand is at 5.

So, the minute hand is 25 minutes or 150° behind the hour hand.

Both Hour and Minute hand coincide are perpendicular to each other if minute hand gains $150^\circ + 90^\circ = 240^\circ$ or $150^\circ - 90^\circ = 60^\circ$ i.e it gains 40 minutes or 10 minutes over the hour hand.

(i) As we know Minute hand gains 55 minutes over the hour hand in 60 minutes.

So, Minute hand gains 1 minute over the hour hand in $\frac{60}{55}$ minutes.

Minute hand gains 40 minutes over the hour hand in $\frac{60}{55} \times 40 = \frac{480}{11} = 43\frac{7}{11}$ minutes.

(ii) As we know Minute hand gains 55 minutes over the hour hand in 60 minutes.

So, Minute hand gains 1 minute over the hour hand in $\frac{60}{55}$ minutes.

Minute hand gains 10 minutes over the hour hand in $\frac{60}{55} \times 10 = \frac{120}{11} = 10\frac{10}{11}$ minutes.

Hence both the hands are perpendicular to each other at 5 hour $10\frac{10}{11}$ minutes and 5 hour $43\frac{7}{11}$

minutes

Q. At what time after 4 O' Clock both Hour and Minute hand are opposite to each other?

Q. At what time between 4 O' Clock and 5 O' Clock both Hour and Minute hand are opposite to each other?

Ans: At 4 O' Clock the minute hand is at 12 and hour hand is at 4.

So, the minute hand is 20 minutes or 120° behind the hour hand.

Both Hour and Minute hand will be opposite to each other if minute hand gains $120^\circ + 180^\circ = 300^\circ$ i.e. 50 minutes over the hour hand.

As we know Minute hand gains 55 minutes over the hour hand in 60 minutes.

So, Minute hand gains 1 minute over the hour hand in $\frac{60}{55}$ minutes.

Minute hand gains 50 minutes over the hour hand in $\frac{60}{55} \times 50 = \frac{600}{11} = 54\frac{6}{11}$ minutes.

Hence both the hands are opposite to each other at 4 hour $54\frac{6}{11}$ minutes.

Q. A person left home between 4 PM and 5 PM and returned home between 5 PM and 6 PM and found that his Hour and Minute hand are changed their places. When did he left the home?

Ans: Hour hand is in x minute division i.e. $20 < x < 25$ and when he returned back then hour hand is at y minute division i.e. $25 < y < 30$.

As minute hand moves 12 times faster than hour hand. Therefore considering the position of the minute hand at the time of going out and returned back we have:

$$y = 12(x - 20) \quad \text{and} \quad x = 12(y - 25)$$

$$\Rightarrow 12x - y = 240 \quad \text{and} \quad x - 12y = -300$$

$$\text{After solving } x = \frac{3180}{143} \quad \text{and} \quad y = \frac{3840}{143} = 26\frac{122}{143}$$

Hence he left the home at 4 hour $26\frac{122}{143}$ minutes.

Exercise:

- 1. At what time after 8 O' Clock both Hour and Minute hand coincide with each other?**
- 2. At what time between 2 O' Clock and 3 O' Clock both Hour and Minute hand coincide with each other?**
- 3. At what time between 2 O' Clock and 3 O' Clock both Hour and Minute hand are perpendicular to each other?**
- 4. At what time between 2 O' Clock and 3 O' Clock both Hour and Minute hand are opposite to each other?**
- 5. A person left home between 2 PM and 3 PM and returned home between 4 PM and 5 PM and found that his Hour and Minute hand are changed their places. When did he left the home?**
